



ENGIN 604 INTRODUCTION TO PYTHON FOR FINANCE

STOCK MARKET DATA USING THE YAHOO FINANCE API

Professor: *Gabriel E. Cabrera*
Teaching Assistant: *Alex Den Braber*



APIs (Application Programming Interfaces) let your products and services communicate with others, without needing to know how they are implemented. A widely used API is the Yahoo Finance one, which lets the user access and download stock market data at different frequencies. In python there is the `yfinance` library that provides access to that API. To install `yfinance`:

```
!pip install yfinance
```

Visit <https://pypi.org/project/yfinance/> to see the library's documentation.

1 Yahoo Finance API

1. Download the stock market data for GameStop (GME) from 2015-01-01 to 2021-03-01 at daily frequency.

```
import pandas as pd # imports pandas
import numpy as np # import numpy
import yfinance as yf # import ytfinance

# ticker
asset = "GME"

# downloads GameStop's OHCL data
gme_ohcl_daily = yf.download(asset, start = "2015-01-01", end = "2021-03-01")

## [*****100%*****] 1 of 1 completed
```

2. Produce brief descriptive statistics including: the total number of observations, mean, standard deviation, minimum, maximum, and the 25th, 50th and 75th percentiles.
3. Extract the year, month and day from the index.
4. Select the closing price (`Close`).

2 Changes Over Time

1. Using a lambda (anonymous) function, generate the difference of the closing price:

$$\Delta_t = p_t - p_{t-1}$$

Where p_t is the closing price at t and p_{t-1} the closing price at p_{t-1} (lag).

2. Using a lambda (anonymous) function, generate the first lag of p_t .

3. Divide the variable created in (1) by the variable created in (2).
4. Use the `pct_change()` method and compare it with the variable created in (3).

3 Basic Time Series Visualization

1. Plot GameStop's closing price.

3.1 Technical Analysis: Moving Average

1. Using GameStop's (GME) closing price, generate:
 - a. The 20-day rolling minimum price.
 - b. The 20-day moving average (short term).
 - c. The 20-day rolling maximum price.

Drop the NAs and plot the variables created from 2020-01-01 to 2020-09-08.

2. To the DataFrame created in (1), add:
 - a. The 252-day moving average (long term).
 - b. A variable equal to 1 if the 20-day moving average > the 252-day moving average, and -1 otherwise.

Drop the NAs and plot the variables created from 2020-01-01 to 2020-09-08.

3. Filter GameStop's OHCL data between 2020-01-01 and 2020-09-08. Then, using the `plotly` library, plot a *Candle-Stick* chart (see <https://plotly.com/python/candlestick-charts/>).

```
# imports plotly
import plotly.graph_objects as go

# filters the OHCL data
part_i = (gme_ohcl_daily.index >= "2020-01-01") # first filter
part_ii = (gme_ohcl_daily.index <= "2020-09-08") # second filter
df_plotly = gme_ohcl_daily[part_i & part_ii]

# plots candlestick
fig_plotly = go.Figure(data=[go.Candlestick(x=df_plotly.index,
                                             open=df_plotly['Open'],
                                             high=df_plotly['High'],
                                             low=df_plotly['Low'],
                                             close=df_plotly['Close'])])

# displays it
fig_plotly.show()
```

4 Downloading Multiple Indices

1. Using the `yfinance` library, download at monthly frequency the OHCL data from 2000-01-01 to 2021-03-01 for the following stocks: Facebook (FB), Amazon (AMZN), Apple (AAPL), Netflix (NFLX) and Google (GOOG).
2. Plot the evolution of prices for all the indices.
3. Reshape the DataFrame generated in (1) so that the columns end up in the rows.
4. Return the previous DataFrame to its original shape.